Northbridge University Endowment — Monthly Risk Report

For the Investment Committee · As of 30 September 2026 · NAV $2.0bn (fictional) · Strategic allocation: EEM 19%, TLT 15%, HYG 16%, VNQ 40%, GLD 10%

Key messages

- One-day 99% VaR is $35.6m (1.78% of NAV) on historical simulation, the model that has been best calibrated since 2010 (41 exceptions vs. 42 expected). Expected Shortfall at 97.5% is $39.8m.
- Current EWMA volatility is 10.6% a year, below the 16.9% realised since 2007: markets are calm, so the EWMA model currently shows the lowest 99% VaR. Its forecasts rise fastest when volatility returns.
- REITs (VNQ) and emerging markets (EEM) are 59% of the portfolio and 66% of its volatility.
- The worst historical stress, the global financial crisis, would cost 38.1% ($761m) on today's weights; the hypothetical recession costs 16.1% ($321m).
- Backtest: the EWMA-normal model failed Kupiec's test (81 exceptions vs. 42) and is in the Basel yellow zone today. All three models fail Christoffersen's independence test: exceptions cluster in stress periods.

1. Value-at-Risk and Expected Shortfall (1 day)

Model	VaR 95%	VaR 99%	ES 97.5%	10-day VaR 99%, √10
Historical simulation (500 days)	$22.1m (1.11%)	$35.6m (1.78%)	$39.8m (1.99%)	5.62%
Parametric EWMA (λ = 0.94)	$21.7m (1.09%)	$30.7m (1.54%)	$30.9m (1.54%)	4.86%
Monte Carlo Student-t (ν = 5)	$20.4m (1.02%)	$33.4m (1.67%)	$34.6m (1.73%)	5.28%

Overlapping 10-day historical returns give a 99% 10-day VaR of 4.58% over the last 500 days and 8.00% over the full sample since 2007; √10 scaling sits between the two.

2. Risk decomposition (EWMA covariance)

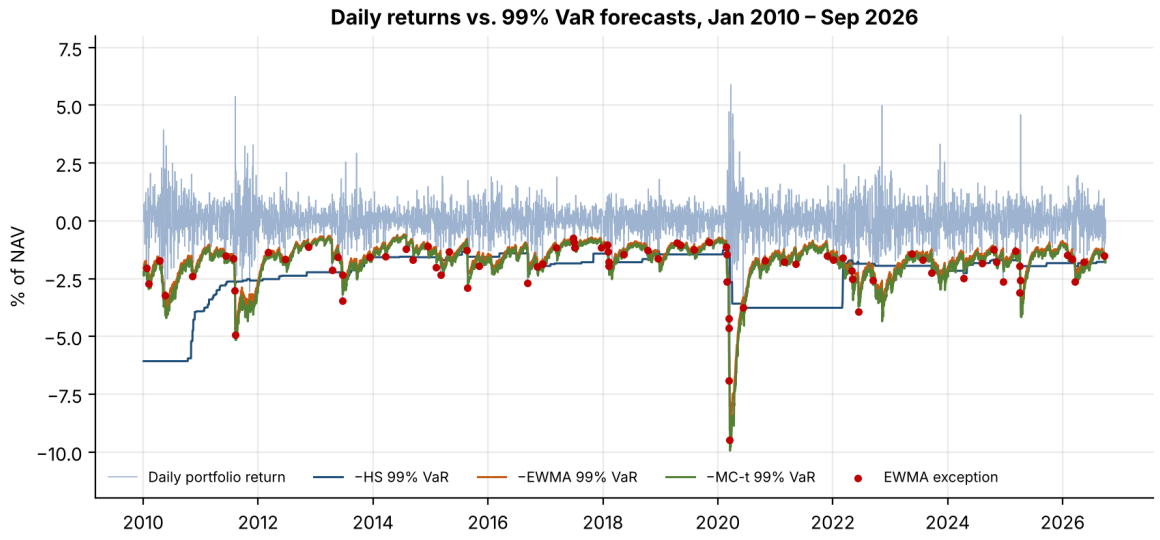
ETF	Weight	Component VaR 99%	Share of volatility	Stand-alone vol.
EEM	19%	$9.5m	30.8%	21.5%
TLT	15%	$4.0m	13.0%	11.3%
HYG	16%	$1.6m	5.3%	4.1%
VNQ	40%	$10.9m	35.2%	11.5%
GLD	10%	$4.8m	15.7%	25.1%

Component VaRs add up to the EWMA 99% VaR for 1 October 2026 (1.55%). Gold's stand-alone volatility is the highest in the portfolio after its recent rally.

3. Backtest of 99% VaR, January 2010 – September 2026

Model	Exceptions / expected	Kupiec p	Christoffersen p	Basel zone now (last 250 days)	Windows green / yellow / red
Historical simulation	41 / 42.1	0.863	0.001	Green (1)	86% / 14% / 0%
Parametric EWMA	81 / 42.1	0.000	0.005	Yellow (5)	52% / 46% / 2%
Monte Carlo Student-t	57 / 42.1	0.029	0.001	Green (3)	81% / 18% / 1%

4,211 trading days. p < 0.05 rejects correct coverage (Kupiec) or independence (Christoffersen).



Source: yfinance adjusted closes; Project 5 policy weights; author's calculations.

4. Stress tests on today's weights

Scenario	Type	Window / shock	Portfolio	P&L
Global financial crisis	Historical	2007-10-09 to 2009-03-09	-38.1%	–$761m
US downgrade and euro crisis	Historical	2011-07-22 to 2011-10-03	-11.1%	–$221m
COVID crash	Historical	2020-02-19 to 2020-03-23	-24.2%	–$484m
Inflation shock	Historical	2022-01-03 to 2022-10-12	-25.7%	–$514m
Recession	Hypothetical	Equities –30%, yields –100 bp, spreads +250 bp	-16.1%	–$321m
Stagflation	Hypothetical	Equities –20%, yields +150 bp, commodities +25%, gold +15%	-13.9%	–$277m

Historical: buy-and-hold ETF total returns between the closes on the start and end dates. Hypothetical: TLT revalued with effective duration 14.63 years and convexity 3.06, HYG with duration 3.25 years, used as both rate and spread duration (ishares.com, as of Oct 05, 2026 and Oct 06, 2026).